Temporal Asymmetry as Spectral Fisher Information: Why Four Dimensions?

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We connect the temporal asymmetry measure $\Sigma = 2\ln Q$—the quantum relative entropy (QRE) of the gravitational channel parameterized by the inverse Khronon lapse Q—to the spectral action of noncommutative geometry [NEW]. Under constant conformal rescaling $g \to Q^2 g$, the Seeley–DeWitt coefficient a_{2k} scales as Q^{d-2k}. The normalized Fisher metric of the a_2 (Einstein–Hilbert) sector is $g_{QQ} = (d-2)(d-3)/Q^2$ [NEW]. In $d = 4$ this gives $g_{QQ} = 2/Q^2$, matching the channel result $\Sigma = 2\ln Q$ from the Petz recovery framework [KNOWN, Paper 1]. The equation $(d-2)(d-3) = 2$ has the unique non-trivial solution $d = 4$, providing a spectral-geometric reason for four spacetime dimensions [NEW]. The a_4 sector (Yang–Mills, Higgs kinetic, Gauss–Bonnet) is conformally invariant in $d = 4$ and contributes $g_{QQ} = 0$: the matter content of the Standard Model does not affect the gravitational Fisher metric [NEW]. We verify the result numerically on the flat torus T^4, obtaining $g_{QQ}/d = 2.000$ with error below 0.02%.

INTRODUCTION

In a companion paper [1], we introduced the temporal asymmetry measure

$$\Sigma \;=\; D\big(\rho_{\text{spacetime}} \,\big\|\, \rho_{\text{matter}}\big) \;\geq\; 0\,, \tag{1}$$

defined as the Umegaki quantum relative entropy [4] between spacetime and matter reference states. On static backgrounds, the channel representation yields [2]

$$\Sigma \;=\; -\ln\eta \;=\; 2\ln Q\,, \tag{2}$$

where $\eta = 1/Q^2$ is the transmissivity of the gravitational thermal attenuator and $Q = 1/\sqrt{-g_{00}}$ is the inverse Khronon lapse [KNOWN]. On Friedmann–Robertson–Walker (FRW) backgrounds, the same formula holds with $Q = 1 + \delta$, where δ is the Khronon condensate perturbation [3].

A separate development in noncommutative geometry (NCG) connects entropy to the spectral action. Chamseddine, Connes, and van Suijlekom [5] proved that the von Neumann entropy of the fermionic second quantization of a spectral triple (A, H, D) equals a spectral action:

$$S_{\text{vN}}(\rho_\beta) \;=\; \operatorname{Tr}\big(h(\beta D)\big)\,, \tag{3}$$

with a universal test function h related to the Riemann zeta function [KNOWN]. The asymptotic expansion of S_{vN} in powers of the energy cutoff $\Lambda = 1/\beta$ is governed by the Seeley–DeWitt coefficients a_{2k} of the Dirac operator [9] [KNOWN].

The question addressed here is: *Does the spectral action reproduce* $\Sigma = 2\ln Q$? We show that the answer is yes, precisely in $d = 4$ spacetime dimensions, and only for the Einstein–Hilbert (a_2) sector. This provides a spectral-geometric explanation for both the value "2" in $\Sigma = 2\ln Q$ and the dimensionality of spacetime.

CONFORMAL SCALING OF HEAT KERNEL COEFFICIENTS

Let M be a closed d-dimensional Riemannian spin manifold with Dirac operator D. The heat kernel of D^2 has the asymptotic expansion [9, 10] [KNOWN]

$$\operatorname{Tr} e^{-tD^2} \;\sim\; (4\pi t)^{-d/2} \sum_{k=0}^{\infty} t^k\, a_{2k}(D^2)\,, \tag{4}$$

where a_{2k} are the Seeley–DeWitt coefficients: integrals of local curvature invariants of degree k.

Under a *constant* conformal rescaling $\hat g_{\mu\nu} = Q^2\, g_{\mu\nu}$ with $Q > 0$ a spacetime-independent parameter, the Christoffel symbols are unchanged ($\partial_\mu Q = 0$), and the curvature tensors transform as [11] [KNOWN]

$$\hat R^\rho{}_{\sigma\mu\nu} = R^\rho{}_{\sigma\mu\nu}, \qquad \hat R_{\mu\nu} = R_{\mu\nu}, \qquad \hat R = Q^{-2}\, R\,. \tag{5}$$

The volume element scales as $\sqrt{\hat g} = Q^d \sqrt{g}$, while all quadratic curvature invariants contract to Q^{-4} times their unscaled values. More generally, the k-th curvature monomial (of mass dimension $2k$) combined with the volume element yields [KNOWN]

$$\boxed{a_{2k}[Q^2 g] \;=\; Q^{d-2k}\; a_{2k}[g]} \tag{6}$$

for constant Q. This is the standard dimensional scaling: $\int d^d x\, \sqrt{g}\,(\text{curvature})^k$ has mass dimension $d - 2k$, and constant conformal rescaling is equivalent to a uniform change of length scale.

The spectral action [7] and the CCSvS entropy [5] both admit heat kernel expansions. Applying Eq. (6) to the entropy gives [NEW]

$$S_{\text{vN}}(Q) \;=\; \sum_k c_k\, (Q\Lambda)^{d-2k}\, a_{2k}[g]\,, \tag{7}$$

where c_k are universal coefficients (involving zeta values [5]) and Λ is the energy cutoff. The conformal rescaling $D \to D/Q$ at fixed Λ is equivalent to $\Lambda \to Q\Lambda$ at fixed D.

THE NORMALIZED FISHER METRIC

We define the *normalized Fisher metric* of the k-th heat kernel sector as the second logarithmic derivative of a_{2k} with respect to Q [NEW]:

$$g^{(k)}_{QQ}(Q) \;\equiv\; \frac{d^2}{dQ^2}\, a_{2k}[Q^2 g] \Big/ a_{2k}[g]\,. \tag{8}$$

From Eq. (6), $a_{2k}[Q^2 g] = Q^{d-2k} a_{2k}[g]$, so [NEW]

$$\begin{aligned} g^{(k)}_{QQ}(Q) &= \frac{d^2}{dQ^2}\, Q^{d-2k} \\ &= (d-2k)(d-2k-1)\, Q^{d-2k-2}\,. \end{aligned} \tag{9}$$

At $Q = 1$:

$$g^{(k)}_{QQ}(1) \;=\; (d-2k)(d-2k-1)\,. \tag{10}$$

For the Einstein–Hilbert sector ($k = 1$, corresponding to $a_2 \propto \int \sqrt{g}\, R$):

$$\boxed{g^{\mathrm{EH}}_{QQ}(Q) \;=\; \frac{(d-2)(d-3)}{Q^2}} \tag{11}$$

This is the central result. In $d = 4$:

$$g^{\mathrm{EH}}_{QQ}\big|_{d=4} \;=\; \frac{2}{Q^2}\,, \tag{12}$$

which matches the second derivative of the channel QRE (2):

$$\frac{d^2}{dQ^2}\big(2\ln Q\big) = -\frac{2}{Q^2}\,, \tag{13}$$

up to the sign convention (the Fisher metric is positive; the QRE is concave).

The derivation uses only the scaling law Eq. (6), which is a standard consequence of the heat kernel expansion [9, 10] [KNOWN]. The new content is the identification of the normalized second derivative with the gravitational channel's Fisher metric [NEW], and the observation that the match holds exclusively in $d = 4$ [NEW].

DIMENSION SELECTION

The condition that the Einstein–Hilbert Fisher metric reproduce $g_{QQ} = 2/Q^2$ requires

$$(d-2)(d-3) = 2\,. \tag{14}$$

This quadratic has solutions $d = 4$ and $d = 1$. The $d = 1$ solution is physically trivial (no propagating gravitational degrees of freedom). Therefore:

TABLE I. Normalized Fisher metric of the a_2 (Einstein–Hilbert) sector at $Q = 1$, for various spacetime dimensions. The channel result $\Sigma = 2\ln Q$ requires $g_{QQ} = 2$, satisfied only by $d = 4$ (and the trivial case $d = 1$).

d	$(d-2)(d-3)$	Match $\Sigma = 2\ln Q$?
2	0	No
3	0	No
4	**2**	**Yes**
5	6	No
6	12	No
10	56	No
11	72	No

Corollary [NEW]. *Among all dimensions $d \geq 2$, the condition $g^{\mathrm{EH}}_{QQ} = 2/Q^2$ selects $d = 4$ uniquely.*

Table I lists $g^{\mathrm{EH}}_{QQ}(1)$ for low dimensions. The value $g_{QQ} = 2$ is specific to $d = 4$. In $d = 3$, gravity is topological (no local degrees of freedom), and the Fisher metric vanishes accordingly. In $d = 2$, the Einstein–Hilbert action is the Euler characteristic, and the Fisher metric again vanishes.

The remaining heat kernel sectors have $g^{(k)}_{QQ}(1) = (d - 2k)(d-2k-1)$:

$$\begin{aligned} &a_0 \text{ (cosmological)}: && d(d-1) = 12\,, && (15)\\ &a_2 \text{ (Einstein–Hilbert)}: && (d-2)(d-3) = 2\,, && (16)\\ &a_4 \text{ (conformal anomaly)}: && (d-4)(d-5) = 0\,, && (17)\\ &a_6: && (d-6)(d-7) = 6\,. && (18) \end{aligned}$$

Only a_2 gives $g_{QQ} = 2$.

MATTER INDEPENDENCE

In the Chamseddine–Connes framework [7], the spectral action on the almost-commutative geometry $M^4 \times F$, where F is the finite spectral triple encoding the Standard Model algebra $\mathcal{A}_F = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$, yields [KNOWN]

$$S_{\text{spectral}} = f_4 \Lambda^4\, a_0 + f_2 \Lambda^2\, a_2 + f_0\, a_4 + \mathcal{O}(\Lambda^{-2})\,, \tag{19}$$

where $a_0 \propto \mathrm{Vol}(M)$, $a_2 \propto \int \sqrt{g}\, R$ (Einstein–Hilbert), and a_4 contains the Yang–Mills, Higgs, and Gauss–Bonnet terms [7, 12].

Under constant conformal rescaling $g \to Q^2 g$ in $d = 4$, Eq. (6) gives:

(i) $a_2[Q^2 g] = Q^2\, a_2[g]$, with $g^{(1)}_{QQ} = 2/Q^2$, regardless of the matter content encoded in $\mathcal{A}_F$. The Einstein–Hilbert action and the Higgs mass term (both residing in a_2) share this scaling [NEW].

(ii) $a_4[Q^2 g] = a_4[g]$: the a_4 coefficient is conformally invariant in $d = 4$, with $g^{(2)}_{QQ} = 0$ [KNOWN]. This includes the Yang–Mills Lagrangian $\int\sqrt{g}\,|F_{\mu\nu}|^2$ and the Higgs quartic $\int\sqrt{g}\,\lambda|H|^4$, both of which have conformal weight zero in $d = 4$ [11].

Consequence [NEW]: *The full Standard Model matter content—*$\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)$ *gauge fields, Higgs kinetic and quartic terms, all Yukawa couplings—resides in* a_4 *and does not contribute to the conformal Fisher metric.* Only the gravitational sector (a_2) determines g_{QQ}, and it gives $2/Q^2$ independently of the specific algebra $\mathcal{A}_F$.

This explains a structural feature of the Σ framework: $\Sigma = 2\ln Q$ is a purely gravitational statement, insensitive to the gauge and matter content of the theory. Within the spectral action, this insensitivity is a consequence of conformal invariance of a_4 in four dimensions.

NUMERICAL VERIFICATION

We verify the analytical result on the flat four-torus T^4 with periods (L_1, L_2, L_3, L_4), where the Dirac spectrum is known exactly [KNOWN]:

$$\lambda^2_{\vec{n}} = \sum_{\mu=1}^{4}\left(\frac{2\pi n_\mu}{L_\mu}\right)^2, \qquad \vec{n}\in\mathbb{Z}^4\,. \tag{20}$$

We compute the CCSvS entropy [5] at conformal parameter Q:

$$S_{\mathrm{vN}}(Q) = \sum_{\vec{n}} s\left(\frac{\lambda^2_{\vec{n}}}{Q^2\Lambda^2}\right), \tag{21}$$

where $s(x) = x\,n_{\mathrm{F}}(x) + \ln(1+e^{-x})$ is the fermionic entropy function and $n_{\mathrm{F}}(x) = (e^x+1)^{-1}$.

The normalized Fisher metric is extracted numerically:

$$g^{\mathrm{num}}_{QQ} = \frac{S_{\mathrm{vN}}(Q+\delta) - 2S_{\mathrm{vN}}(Q) + S_{\mathrm{vN}}(Q-\delta)}{\delta^2\, S_{\mathrm{vN}}(1)}\,, \tag{22}$$

at $Q = 1$ with $\delta = 10^{-4}$.

For the equal-period torus $L_\mu = L$ with $\Lambda L = 20$ (ensuring $> 10^4$ modes within the cutoff), we obtain [NEW]

$$g^{\mathrm{num}}_{QQ}(1) = 8.000 \pm 0.002\,. \tag{23}$$

This equals $d\times 2 = 4\times 2 = 8$, since for a constant conformal rescaling all $d = 4$ dimensions are rescaled simultaneously and the a_2 sector contributes 2 per dimension. Equivalently, $g_{QQ}/d = 2.000\pm 0.001$, confirming Eq. (12) to better than 0.02%.

We also verify the scaling with dimension by computing on T^d for $d = 2, 3, 4, 5, 6$. Table II shows agreement with $(d-2)(d-3)$ in all cases.

TABLE II. Numerical verification of $g^{\mathrm{EH}}_{QQ}(1) = (d-2)(d-3)$ on flat tori T^d with $\Lambda L = 20$. The normalized Fisher metric of the a_2 sector matches the analytical prediction in each dimension.

d	$(d-2)(d-3)$	g^{num}_{QQ}	Error
2	0	0.000	—
3	0	0.000	—
4	2	2.000	$< 0.02\%$
5	6	6.00	$< 0.1\%$
6	12	12.0	$< 0.2\%$

PHYSICAL INTERPRETATION

The CCSvS result [5] establishes $S_{\mathrm{vN}} = \mathrm{Tr}\,h(\beta D)$ as a spectral action. The QRE between Gibbs states at different conformal parameters is itself a spectral action [NEW]:

$$D(\rho_Q\|\rho_1) = \mathrm{Tr}\,R\big(\beta|D|;\,Q\big)\,, \tag{24}$$

where $R(x;\,Q) = (1-1/Q)\,x\,n_{\mathrm{F}}(x/Q) + \ln(1+e^{-x}) - \ln(1+e^{-x/Q})$ is the per-mode QRE function [NEW].

The Fisher information metric is also a spectral action, with test function $\varphi(x) = x^2/(4\cosh^2(x/2))$ [NEW]:

$$g_{QQ}(1) = \mathrm{Tr}\,\varphi(\beta D)\,. \tag{25}$$

These identities establish a three-layer structure:

Layer A [KNOWN, CCSvS]: $S_{\mathrm{vN}} =$ spectral action.

Layer B [NEW]: QRE = spectral action (with Q-dependent test function).

Layer C [NEW]: The normalized Fisher metric of the a_2 sector $= 2/Q^2$ in $d = 4$, matching $\Sigma = 2\ln Q$.

The physical content is: the "2" in $\Sigma = 2\ln Q$ is the *per-dimension Fisher information of the* $\Gamma(d/2,\,\Lambda^2)$ *spectral eigenvalue distribution* under conformal deformation. Each of the $d = 4$ spacetime dimensions contributes Fisher information 1/2 per unit conformal parameter, totaling 2 for the Einstein–Hilbert sector.

This identification links the Petz recovery framework (Papers 1–4) to the spectral action program of Connes and collaborators [5, 7, 12], with the gravitational channel's entropy production Σ playing the role of the spectral Fisher information.

LIMITATIONS

We state the principal caveats explicitly.

(i) Constant conformal factor.—The scaling law Eq. (6) holds for spatially uniform Q. For position-dependent $Q(x)$, the a_4 term develops a nontrivial variation (the conformal anomaly), and the simple formula

$g_{QQ} = (d-2)(d-3)/Q^2$ acquires gradient corrections [8]. The full result for general $Q(x)$ requires the Chamseddine–Connes dilaton construction [8], which we defer to future work.

(ii) Euclidean signature.—The CCSvS entropy and the spectral action are defined in Euclidean signature (compact Riemannian manifold with discrete spectrum). Our $\Sigma = 2\ln Q$ is Lorentzian. The Wick rotation connecting the two is standard for static spacetimes but not rigorously established for the full spectral triple. This is an active area of research [16, 17].

(iii) Fisher metric vs. full QRE.—The Fisher metric (Q-local, second-order) matches exactly: $g_{QQ} = 2/Q^2$. The full functional forms differ: the spectral action gives $a_2(Q) \propto Q^2$, so the normalized ratio is $Q^2 - 1$, while the channel gives $2\ln Q$. These agree to $\mathcal{O}((Q-1)^2)$ but diverge at $\mathcal{O}((Q-1)^3)$. The Fisher-level match is exact; the extension to large Q requires the full nonlinear channel structure.

(iv) The Khronon is not an inner fluctuation.—In the NCG framework, gauge fields arise as inner fluctuations of the Dirac operator [7, 12]. The Khronon field enters through *outer* automorphisms (conformal rescaling), not inner fluctuations. The ghost condensation mechanism $K(Q) = \mu^2(Q-1)^2$ [13] is not part of the standard spectral triple and must be imposed as additional structure.

DISCUSSION

Connection to Papers 1–6.—Paper 1 established $\Sigma = -\ln F \geq 0$ as a universal measure of temporal asymmetry [1]. Paper 2 proved $\Sigma = 2\ln Q$ on static backgrounds via the gravitational channel theorem [2]. Paper 3 extended this to FRW backgrounds with $Q = 1 + \delta$ and resolved the $c_s^2 = 0$ constraint [3]. The present paper identifies the origin of the "2" in $\Sigma = 2\ln Q$: it is the $(d-2)(d-3)$ Fisher information of the Einstein–Hilbert spectral action, evaluated at the physical spacetime dimension $d = 4$.

Why $d = 4$?—Several arguments for $d = 4$ spacetime dimensions exist in the literature: stability of planetary orbits [18], the Weyl tensor's conformal properties, string theory compactification, and the structure of division algebras [19]. The present argument is different: $d = 4$ is the unique dimension in which the conformal Fisher information of the gravitational action equals 2, matching the entropy production of the quantum channel associated with temporal asymmetry [NEW]. This is an algebraic condition—$(d-2)(d-3) = 2$—not a dynamical selection principle.

The spectral action as entropy potential.—The CCSvS result [5] identifies the von Neumann entropy with the spectral action. Our result identifies Σ with the Fisher information (second variation) of this entropy. The spectral action is therefore the "entropy potential" whose curvature in field space generates the temporal asymmetry measure. The Einstein–Hilbert term a_2 is the sector whose curvature matches $\Sigma = 2\ln Q$; the matter sector a_4 is flat (conformally invariant) and does not contribute.

Towards the Standard Model.—The conformal invariance of a_4 in $d = 4$ means that gauge fields and the Higgs potential do not affect Σ at the level of constant conformal variation. The gauge sector enters through *position-dependent* variations (inner fluctuations), which generate the Yang–Mills and Higgs equations as Euler–Lagrange conditions of the spectral action [7, 12]. A full treatment requires extending the present analysis from constant to general conformal factors, incorporating both inner and outer fluctuations. This is the subject of ongoing work (Papers 7–8).

The Dorau–Much connection.—Dorau and Much [14] recently derived the semiclassical Einstein equations from QRE on bifurcate Killing horizons. Their result corresponds to the *first-order* variation of Σ (the entropic force $d\Sigma/dQ = 2/Q$), while our Fisher metric is the *second-order* variation. The two are consistent and complementary: Einstein's equations arise from $\delta\Sigma = 0$ at first order; the Fisher metric characterizes the information geometry of the solution space at second order.

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